

The aim of this tutorial is to familiarise you with propositional logic for knowledge-representation and reasoning; that is, how propositional logic can be used to represent knowledge and be used to make inferences. **The ultimate goal is that you are able to use SAT technology to solve problems, by encoding them as a SAT task. This is done in the last part of this tutorial sheet.**

You are not expected to complete all problems in the tutorial session, and you can always catch up and complete unsolved problems later.

A propositional *sentence* or *formula* is:

- A propositional letter, often written as p or q .
- $\neg P$ where P is a sentence.
- $P \wedge Q$ where P and Q are sentences.
- $P \vee Q$ where P and Q are sentences.
- $P \Rightarrow Q$ where P and Q are sentences.
- $P \Leftrightarrow Q$ where P and Q are sentences.

Sometimes we also define *exclusive or* as $P \oplus Q$ where P and Q are sentences.

We refer to $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$ and $\oplus$ as *logical connectives*. In some places, you will find $\rightarrow$ or $\supset$ used instead of $\Rightarrow$ for implication and $\equiv$ instead of $\Leftrightarrow$. In those cases, the symbol $\Leftrightarrow$ is generally used for logical equivalence (i.e., two formulas that have exactly the same truth table).

An *interpretation* in propositional logic is a *truth assignment* (or *valuation*) mapping each atomic propositions to either true or false.

A *truth table* shows the truth value of formulas of interest per possible interpretation of the domain. The truth table showing the semantics (i.e., meaning) of each of the above logical connectives is as follows:

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$	$P \oplus Q$
T	T	F	T	T	T	T	F
T	F	F	F	T	F	F	T
F	T	T	F	T	T	F	T
F	F	T	F	F	T	T	F

A propositional formula is:

- **Satisfiable:** if there is at least one truth assignment that makes it true.
- **Valid or Tautology:** if every truth assignment makes it true. A formula that is valid is also satisfiable.
- **Unsatisfiable:** if there is no truth assignment that makes it true (so that all truth assignments make it false).

Two formulas P and Q are *logically equivalent* (denoted $P \equiv Q$) iff $P \Rightarrow Q$ and $Q \Rightarrow P$ (and hence $P \Leftrightarrow Q$) are tautologies. It means their truth value coincide on every interpretation (i.e., they have the same truth table). For example $P \vee Q$ is logically equivalent to $Q \vee P$ and to $\neg(\neg P \wedge \neg Q)$.

PART 1: Basic Logic

- (a) Use truth tables to show that the following are valid (i.e. that the equivalences hold).

$P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$	Distribution of $\wedge$
$\neg(P \wedge Q) \Leftrightarrow (\neg P \vee \neg Q)$	de Morgan's Law
$\neg(P \vee Q) \Leftrightarrow (\neg P \wedge \neg Q)$	de Morgan's Law
$(P \Rightarrow Q) \Leftrightarrow (\neg Q \Rightarrow \neg P)$	Contraposition
$(P \Rightarrow Q) \Leftrightarrow (\neg P \vee Q)$	

- (b) For each of the following sentences, decide whether it is
- valid**
- ,
- unsatisfiable**
- , or
- satisfiable but not valid**
- . Firstly, trying “guessing” the answer; then evaluate each properly (e.g. using truth tables). How did your guesses match up?

- i. $Smoke \Rightarrow Smoke$
- ii. $Smoke \Rightarrow Fire$
- iii. $(Smoke \Rightarrow Fire) \Rightarrow (\neg Smoke \Rightarrow \neg Fire)$
- iv. $Smoke \vee Fire \vee \neg Fire$
- v. $((Smoke \wedge Heat) \Rightarrow Fire) \Leftrightarrow ((Smoke \Rightarrow Fire) \vee (Heat \Rightarrow Fire))$
- vi. $(Smoke \Rightarrow Fire) \Rightarrow ((Smoke \wedge Heat) \Rightarrow Fire)$
- vii. $Big \vee Dumb \vee (Big \Rightarrow Dumb)$
- viii. $(Big \wedge Dumb) \vee \neg Dumb$

- (c) Consider the natural language statement “If
- A
- then
- B
- else
- C
- ” (in other words, if
- A
- is true then
- B
- is true, and otherwise
- C
- is true). Give two different formulas in propositional logic that capture that statement formally.

- (d) Prove that
- $\neg(P \Rightarrow Q)$
- and
- $\neg(\neg Q \Rightarrow \neg P)$
- are all equivalent without using a truth-table

- (e) Represent the following sentences in propositional logic. Can you prove that the unicorn is mythical? What about magical? Horned?

If the unicorn is mythical, then it is immortal, but if it is not mythical, then it is a mortal mammal.

If the unicorn is either immortal or a mammal, then it is horned. The unicorn is magical if it is horned.

PART 2: Resolution

First, let us go over an example in full. The resolution method can be used to prove that a CNF formula (a set of clauses) is unsatisfiable, by deriving the empty clause. Consider, for example, the following CNF formula:

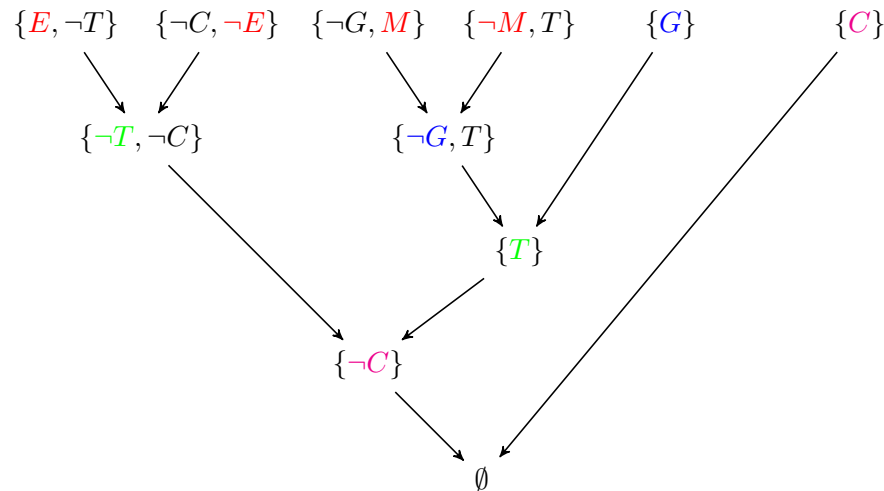
$$\phi = (\neg T \vee E) \wedge (T \vee \neg E) \wedge (M \vee \neg G) \wedge (\neg E \vee \neg C) \wedge (T \vee \neg M) \wedge G \wedge C$$

Here is a resolution proof for the empty clause:

1. $(M \vee \neg G)$ clause in ϕ
2. $(T \vee \neg M)$ clause in ϕ
3. $(\neg G \vee T)$ resolution 1,2
4. $(\neg T \vee E)$ clause in ϕ
5. $(\neg G \vee E)$ resolution 3,4
6. $(\neg E \vee \neg C)$ clause in ϕ
7. $(\neg G \vee \neg C)$ resolution 5,6
8. G clause in ϕ
9. C clause in ϕ
10. $(\neg C)$ resolution 7,8
11. $()$ resolution 9,10

Note that clause $(T \vee \neg E)$ in ϕ was never used, so even if we drop such clause from ϕ , it still remains unsatisfiable. Also observe that, in many cases, there could be more than one (correct) resolution proof deriving the empty clause.

Another way of showing a resolution proof is using a graphical representation of the proof and using sets of literals to denote clauses:



(Colours are added to show the order in which the resolutions occur: red $\Rightarrow$ blue $\Rightarrow$ green $\Rightarrow$ magenta).

OK, now, let's do some exercises....

(a) Use propositional resolution to prove that the following sets of clauses are unsatisfiable. Observe how clauses are now represented via set notation!

- $\{\{p, q\}, \{\neg p, r\}, \{\neg p, \neg r\}, \{p, \neg q\}\}$.
- $\{\{p, q\}, \{\neg r, s\}, \{\neg p, r, s\}, \{\neg q, \neg r\}, \{p, \neg s\}, \{\neg p, \neg r\}, \{r\}\}$.

(b) Show that the following statements are true by giving an appropriate resolution proof. You will need to convert formulas to CNF before doing a resolution proof.

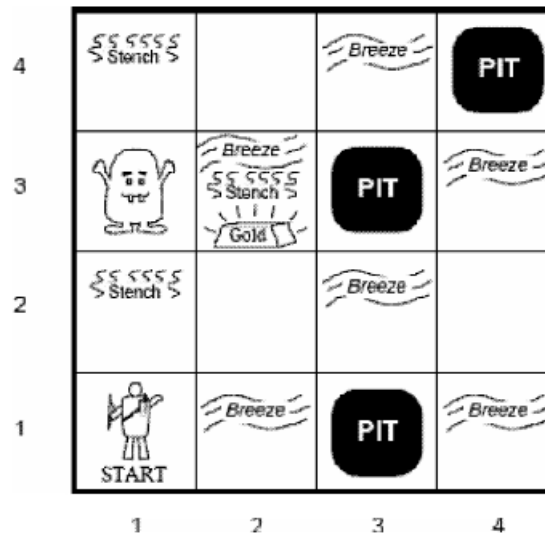
- $p \wedge (p \Rightarrow q) \models q$
- $p \wedge (p \Rightarrow q) \wedge (q \Rightarrow r) \models r$
- $(p \Rightarrow (q \Rightarrow r)) \wedge (r \Rightarrow s) \wedge p \models \neg q \vee s$
- $(p \vee r) \wedge (p \Rightarrow (q \vee s)) \wedge (r \Rightarrow t) \models q \vee s \vee t$
- $p \wedge r \wedge (p \Rightarrow q) \wedge ((q \wedge r) \Rightarrow (s \vee t)) \wedge \neg s \models t$

(c) Complete the truth table below. What does this tell you about resolution proofs?

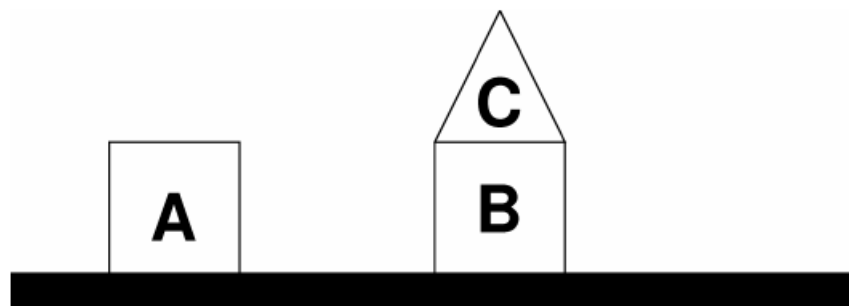
P	Q	R	$\neg P \vee Q$	$P \vee R$	$(\neg P \vee Q) \wedge (P \vee R)$	$Q \vee R$	$((\neg P \vee Q) \wedge (P \vee R)) \Rightarrow (Q \vee R)$
T	T	T	_____	_____	_____	_____	_____
T	T	F	_____	_____	_____	_____	_____
T	F	T	_____	_____	_____	_____	_____
T	F	F	_____	_____	_____	_____	_____
F	T	T	_____	_____	_____	_____	_____
F	T	F	_____	_____	_____	_____	_____
F	F	T	_____	_____	_____	_____	_____
F	F	F	_____	_____	_____	_____	_____

PART 3: Knowledge representation.....

(a) For the following Wumpus world:



- i. Develop a notation capturing the important propositions.
 - ii. How would you express in a propositional logic sentence:
 - α) If square $[2, 2]$ has no smell then the Wumpus is not in this square or any of the adjacent squares?
 - β) If there is stench in square $[1, 2]$ there must be a Wumpus in this square or any of the adjacent squares?
 - iii. Given that the agent is standing in square $[1,1]$ and can see (and smell) all adjacent squares, including diagonals; ; how can the agent deduce that the Wumpus is in square $[1, 3]$ using the laws of inference in propositional logic and the formulas constructed in the previous question.
 - iv. How many propositions are required to fully capture the world in a single formula? What about a general world of size n ?
- (b) Represent the following “blocks world” scene in propositional calculus.



(c) Consider the following information available in the domain of thing encountered at sea:

1. None of the unnoticed things are mermaids.
2. Anything entered into the log is sure to be worth remembering.
3. I have never seen anything worth remembering when on a voyage.
4. Anything that is noticed is sure to be recorded in the log.

Formalise the problem in propositional logic and use resolution to formally prove that *I have never seen a mermaid at sea* using the following atomic propositions:

L : entered into the log
 M : is a mermaid
 S : seen by me
 N : noticed
 W : worth remembering

PART 4: Knowledge bases and resolution

- (a) Consider a knowledge base built of just these three weird implications:

$$\begin{aligned}
 \neg A &\Rightarrow B \\
 B &\Rightarrow A \\
 A &\Rightarrow (C \wedge D)
 \end{aligned}$$

- i. Prove formula $A \wedge C \wedge D$ using Modus Ponens only, or explain why this is not possible.
 - ii. Prove formula $A \wedge C \wedge D$ using resolution.
- (b) Given the following symbols and sentences:
- C to indicate that Gianni is a climber;
 F to indicate that Gianni is fit; L to indicate that Gianni is lucky;
 E to indicate that Gianni climbs mount Everest.
- i. Formalize the above sentences in propositional logic:

If Gianni is a climber and he is fit, he climbs mount Everest.
 If Gianni is not lucky and he is not fit, he does not climb mount Everest.
 Gianni is fit.
 - ii. Tell if the KB built in above is consistent, and tell if some of the following sets are models for the above sentences:
 $\{\}$; $\{C, L\}$; $\{L, E\}$; $\{F, C, E\}$; $\{L, F, E\}$.

(Recall that for the binary variables A , B and C ; the set $S = \{A, C\}$ means A and C are true, and B is false. S is said to be a model of KB iff all statements in KB are true for that given assignment of the variables)
- (c) Prove the propositional formula $[(A \Rightarrow C) \vee (B \Rightarrow C)] \Rightarrow [(A \wedge B) \Rightarrow C]$ is valid using resolution.
- (d) Let A , B , C be propositional symbols. Given $KB = \{A \Rightarrow C, B \Rightarrow C, A \vee B\}$, tell whether C can be derived from KB or not. Use resolution.
- (e) Heads, I win. Tails, you lose. Use propositional resolution to prove that I always win.

PART 5: SAT Technology

Satisfiability in propositional logic (named SAT) is a very active area of research and used in the industry, for constraint satisfaction, optimisation, planning, etc.

- (a) Research and explain what is the DIMACS format used as a standard for specifying CNF formulas (and used by most SAT solvers). Check, for example [this](#) and [this](#).
- (b) Get a SAT solver (e.g., [MiniSAT SAT](#); [SAT4j](#), file `sat4j.jar`; or Microsoft [Z3 Prover](#)), and test it on some CNF files in DIMACS format from [here](#).
- (c) Finally, solve Part 3.c and ALL the problems in PART 4 by making use of a SAT solver.

End of tutorial worksheet